

FLAVORATORY ❀❀

*A Culinary Journey by Lionel Hegetschweiler, Naël Meter &
Arnaud Favre*



THE TASTING MENU: PROCESS BOOK

Appetizers

Path to the final result – The Journey of Flavor

Main Courses

Challenges – Evolution of the Kitchen Plans

The House Service

Peer Assessment – The Role of the Brigade

THE JOURNEY OF FLAVOR

Every great dish begins with raw ingredients, and our project began with a massive, unrefined dataset: thousands of culinary recipes and their underlying chemical flavor compounds (sourced from FlavorDB and the Ahn et al. research). Our primary objective was to demystify the "art" of cooking by exposing the mathematical "science" behind it.

To prepare our foundational data layer, we integrated the raw databases of FlavorDB and CulinaryDB, cross-referencing global recipes with the distinct volatile molecules that give each ingredient its signature taste profile. Once this consolidated matrix was established, we entered an exploratory data analysis phase to test the boundaries of our dataset and discover exactly what visual narratives could be brought to life from the information.

From this analysis, we deduced that a single, monolithic dashboard would completely overwhelm the user. Instead, we structured the visualization framework as a clear, three-stage narrative journey:

1. **The Science (Network Explorer):** We first needed to establish the foundational rule that ingredients pair well because they share molecular structures.
2. **The Sociology (Cultural Maps):** Once the user understood the chemistry, we applied it to geographic data to visualize the "Food Pairing Hypothesis"—proving that Western cuisines gravitate toward chemical overlap, while Eastern cuisines avoid it.
3. **The Application (Pairing Oracle):** Finally, we transitioned from passive storytelling to active exploration, giving the user an interactive tool to generate their own science-based recipes.

With the system architecture and core analytical mechanics implemented across all three visualizations, we shifted our focus to the final user interface polish. We iteratively refined our design stripping away visual noise, introducing modern layout containers, softening transitions, and optimizing spatial alignments. This final design layer bridged the gap between academic chemistry and intuitive user exploration, transforming a dense matrix of information into a highly interactive website.

EVOLUTION OF THE KITCHEN PLANS

Every great dish begins with an ambitious kitchen blueprint. Translating raw chemistry data into a seamless, interactive user experience required careful menu alterations. Below is the breakdown of how our original design sketches evolved into the final visualizations on our website.

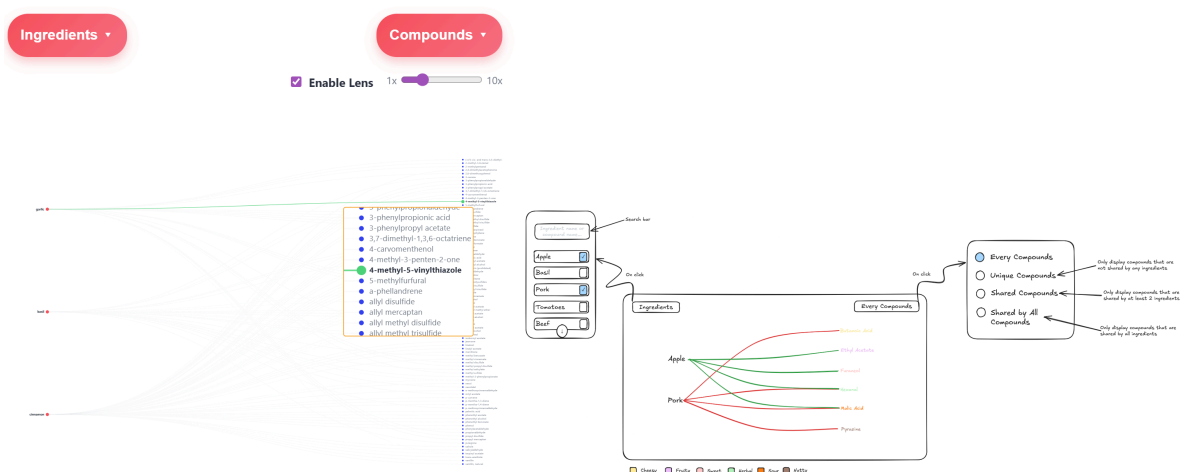
Course 1: Network Explorer (The Bipartite Graph)

This graph was created with the intent of clearly setting the goal of the website and as a tool to complement later visualizations. Using two dropdowns, the users can easily analyze the compatibility of any ingredients and understand why. While the **Cultural Map** and the **Pairing Oracle** serve as our menu mapping out ingredients that

complement or contrast each other, the **Network Explorer** functions as the culinary dictionary, revealing details about each ingredient and explaining the relationship between ingredients.

However, expanding the network introduced a major obstacle regarding legibility. As more ingredients are added, the compound list on the right side grows in size. Because our objective was to maintain a compact layout contained entirely on one screen without scrolling, the text size progressively decreases with each new entry. Overcoming this layout constraint proved to be our greatest design challenge. We faced a choice between two solutions: eliminate the user's ability to switch between compound filter configurations (all, unique, shared, or all-shared) to restrict data density, or develop a feature to enhance readability.

The first option was not optimal. First, it stripped away key analytical features. Secondly, what happens if we select two ingredients close compound wise (such as peanut and peanut butter) and we only allow the user to see shared compounds? Well, we get the exact same situation we wanted to avoid earlier. So we came up with a new idea: adding a magnifying lens. This allows the user to expand microscopic elements on hover while keeping the core functionality and the container size intact.



Upon inspecting the refined interface, located just beneath the compound filtering controls, we integrated an interactive toggle for the **Magnifying Lens** alongside a zoom adjustment slider. This dynamic overlay follows the cursor making it highly intuitive to use and ease the navigation of dense data.

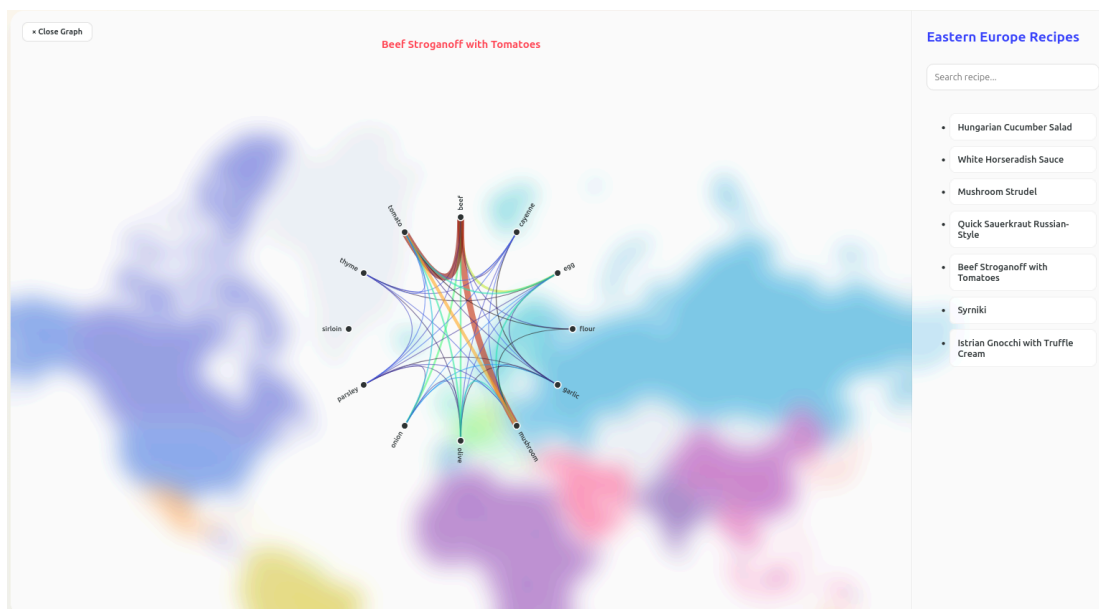
Another change involved the compound categorization. Due to constraints in raw chemical metadata, this feature was discarded. However, we concluded that the core analytical power of the visualization remains fully intact, successfully preserving our primary objective.

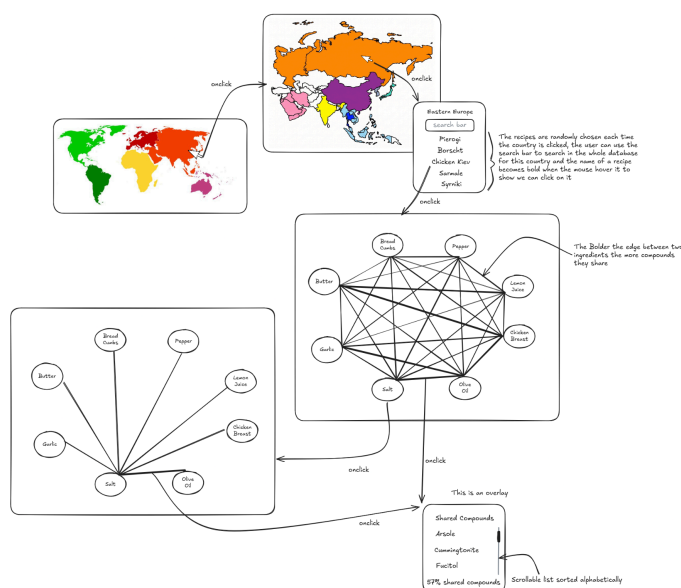
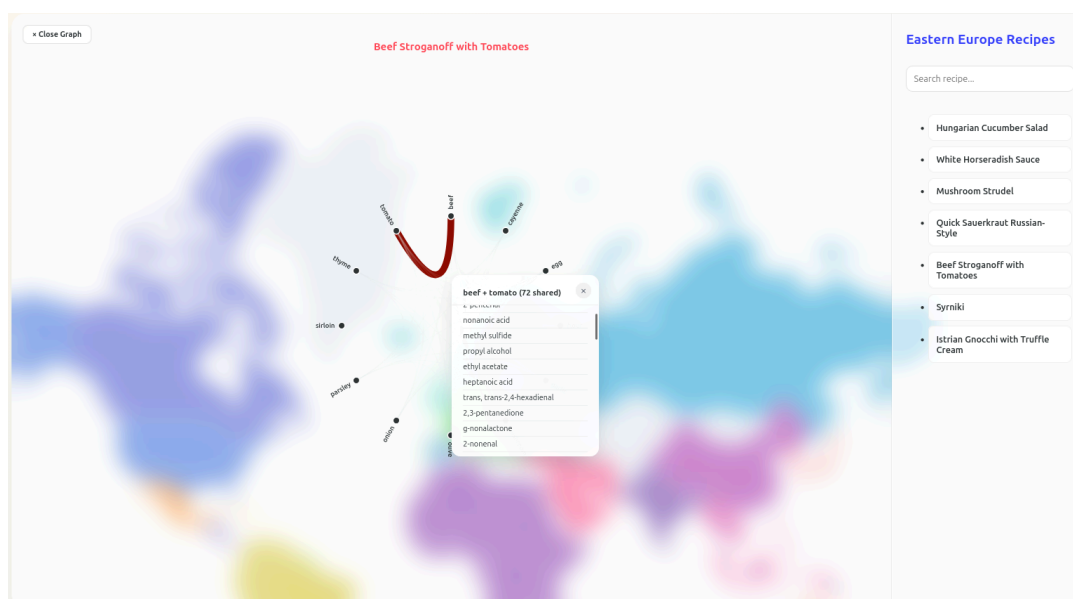
Course 2: Cultural Map (The Interactive World Map & Recipe Sub-Graphs)

Conceived as a gateway to global gastronomy, this visualization allows users to navigate the world map to uncover ingredient use regionally. It serves as an analytical lens for distinguishing between the *Western harmony* of the Food Pairing Hypothesis and the *culinary variance* found in the contrast principles of other geographic regions.

The interface invites users to select a highlighted country group to unroll localized recipes. Upon selecting a specific dish, the canvas displays a **Circular Connection Graph** where ingredients act as nodes and where edge stroke thickness directly represents the number of shared compounds between two ingredients.

Our primary design evolution involved abandoning the original plan to have a floating card for the recipes. Instead, we created a side-by-side split screen interface. This ensures the user can switch between countries and recipes quickly.



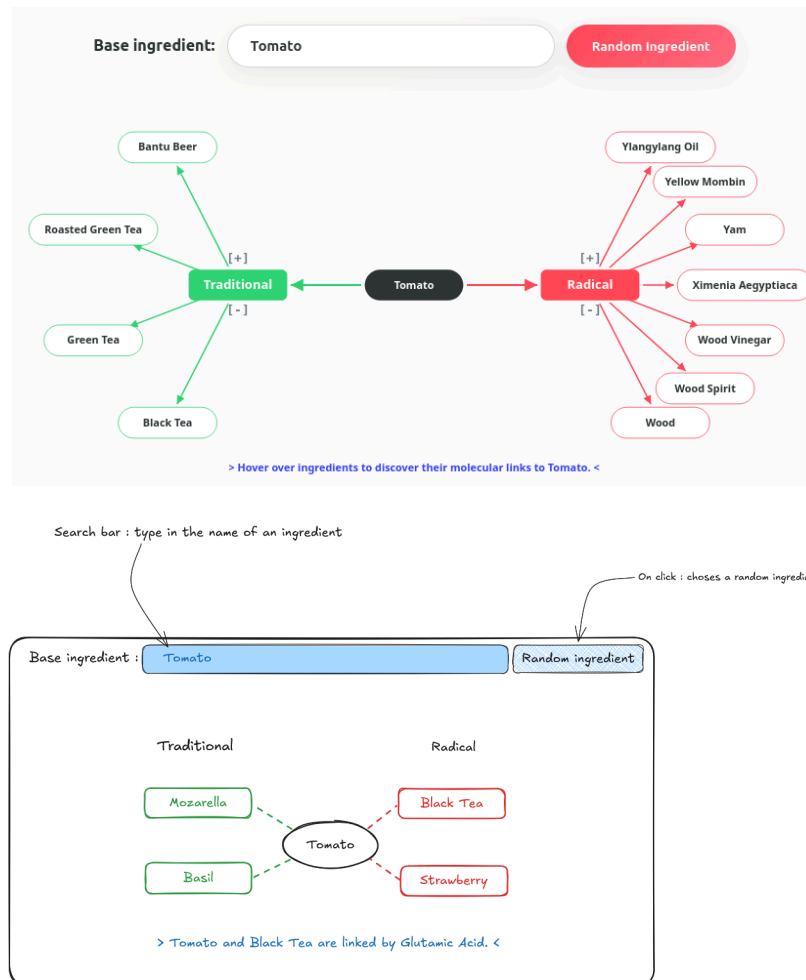


Beyond this layout refinement, the core analytical features were preserved, including the dynamic **Edge Filtering** and the descriptive overlay that identifies the specific shared molecules bridging each ingredient pair.

Course 3: Pairing Oracle (Ingredients Search Bar)

Conceived as the final, actionable *digestif* of our narrative menu, the Pairing Oracle transitions the user from a passive observer of global gastronomy into an active flavor alchemist. While previous courses mapped out existing culinary data, the Oracle functions as a dynamic invention workstation, allowing chefs and home cooks to input a single base ingredient and discover sound flavor combinations based on shared chemistry.

The algorithm dynamically segregates recommendations into two distinct paths: **Traditional Pairings** (ingredients that share an abundance of chemical compounds) and **Radical Pairings** (highly unconventional choices that share few or no compounds but offer a contrasting masterpiece).



This visualization didn't go under heavy modifications compared to the original sketch apart from the fact that the user can change the number of displayed ingredients in the Traditional and Radical category.

THE ROLE OF THE BRIGADE

The Role of the Brigade

In our kitchen, there is no traditional vertical hierarchy. We operated as three Co-Executive Chefs, splitting the workload equally. Every member brought their unique expertise to the table, continuously reviewing and refining each other's work in a collaborative environment to ensure a perfectly balanced final menu.

- **Lionel Hegetschweiler | Co-Executive Chef**

- **The First Course & Narrative Seasoning (Tasks):** Prepared Visualization 1 (Network Explorer), setting the complex technical foundation of the flavor network, and crafted the storytelling on the website to guide the patron's journey.
- **Peer Assessment:** Lionel successfully built the project's technical foundation and narrative structure. His peers commended his ability to translate dense chemistry data into an engaging visualization and story alongside it.

- **Arnaud Favre | Co-Executive Chef**

- **The Main Course & Visual Presentation (Tasks):** Conceived and implemented Visualization 2 (Cultural Maps), building the complex interactive bridge between geographic data and dynamic recipe sub-graphs.
- **Peer Assessment:** Arnaud managed the complex integration of geographic data and dynamic network graphs. The team highly valued his focus on UI/UX, creating a clean interface that presents heavy spatial and relational information without overwhelming the user, and the application of his problem-solving abilities to the website as a whole as visualization 2 is a hard one.

- **Naël Meter | Co-Executive Chef**

- **The Finishing Plating & Innovation (Tasks):** Elegantly engineered Visualization 3 (Pairing Oracle), serving as the final, refined digestif to the culinary experience.
- **Peer Assessment:** Naël's clean approach was heavily praised by his peers. He successfully transformed abstract mathematical concepts into an intuitive and playful tool that encourages users to experiment with the data. He also heavily participated in the screencast and the process book.